

ANOVA intro, part 1

In this video, you will learn

- The purpose of an ANOVA.
- How to talk about ANOVAs at stats parties.
- How to determine whether an ANOVA is appropriate for a given research question or dataset.

At its most basic level, variance analysis compares

- **"what happened"**

against

- **"what was expected."**

Expectations are usually set via a budget, a forecast, or past historical data.

Who or what or where is an “ANOVA”?!?

ANalysis Of VAriance

The goal of an ANOVA is to determine the effects of discrete independent variables (groups, levels) on a continuous dependent variable.

Quick example: Effects of medication type and age group on medication treatment.

ANOVA is the quiet hero behind countless research studies and data-driven decisions. It is like a referee in a match impartial, logical, and precise. It tells us whether the differences we observe are real or simply products of chance.

Once you understand its core logic, ANOVA becomes more than just an equation. It becomes a mindset a way to think critically about variation, evidence, and truth in the noisy world of data.

It provides a disciplined way to separate signal from noise, truth from coincidence. Rather than trusting gut instinct, you get statistical evidence that supports your conclusions.

How ANOVA Works in Simple Terms

The logic behind ANOVA is beautifully simple once you look at it conceptually. It separates the total variation in your data into two parts:

1. **Variation between groups** — This measures how different the group means are from each other. If the treatments or conditions genuinely affect the outcome, this variation will be large.
2. **Variation within groups** — This measures how much individuals differ within the same group, which often happens due to natural randomness or measurement error.

If the between-group variation is much greater than the within-group variation, it signals that the groups are likely different in a meaningful way.

This relationship is summarized in a value called the F-statistic. The F-statistic is a ratio it compares the variability between groups to the variability within groups.

If the F-statistic is large, it suggests that at least one group mean is significantly different. If it is small, the differences you see are probably due to chance.

Interpreting ANOVA Results

When you perform an ANOVA test, you usually get three key numbers:

- 1. F-statistic** — The main test statistic that tells you whether the differences between groups are meaningful.
- 2. p-value** — This tells you how likely it is that the observed differences happened by chance.
- 3. Significance level (α)** — A threshold, often 0.05, used to decide whether the p-value is small enough to call the result significant.

If the p-value is less than 0.05, you can conclude that at least one group mean is different. However, ANOVA does not tell you which specific groups differ only that there is a difference somewhere. To find out where, you use post-hoc tests, such as Tukey's test, to pinpoint the exact differences.

Setting up an ANOVA in four steps

- Step 1: Review the experiment design and make sure ANOVA is the appropriate method.
- Step 2: Identify the independent and dependent variables.
- Step 3: Create a table of factors and levels (when possible).
- Step 4: Compute the model and interpret the results.

Setting up an ANOVA in four steps

Step 1: Review the experiment design and make sure ANOVA is the appropriate method.

Research goal: Test whether Covid-19 medications are effective for different groups.

Experiment: Randomly assign patients to receive medication A, B, or placebo. Measure disease severity after 10 days. Separate older (>50 years) from younger (<50 years) patients.

Setting up an ANOVA in four steps

Step 2: Identify the independent and dependent variables.

IVs: Medication and age group.

DV: Disease severity after 10 days.

Setting up an ANOVA in four steps

Step 3: Create a table of factors and levels (when possible).

Factors: The “dimensions” of IVs.

Levels: The specific groups or manipulations within each factor.

Factors: Medication, age.

Levels: A, B, placebo (medication); younger, older (age)

Setting up an ANOVA in four steps

Step 3: Create a table of factors and levels (when possible).

**Note: Not THE ANOVA table*

Factors: Medication, age.

Levels: A, B, placebo (medication); younger, older (age)

		Medication		
		A	B	placebo
Age group	Younger			
	Older			

Setting up an ANOVA in four steps

Step 4: Compute the model and interpret the results.

Main effect: One factor influences the DV even when ignoring all other factors.

Interactions: The effect of one factor depends on the levels of another factor.

Intercept: The average DV is different from zero. The intercept of the ANOVA is usually ignored.

Setting up an ANOVA in four steps

Step 4: Compute the model and interpret the results.

Main effect: Young people's symptoms improve faster than older people's symptoms, regardless of medication type.

Interaction: Medication A works better in older people; medication B works better in younger people.

Intercept: Symptoms improve for almost everyone after 10 days.

Examples of when ANOVAs are inappropriate

Research goal: Test whether people with more Facebook friends have higher self-reported extraversion.

Variables: DV: number of Facebook friends. IV: scores on a personality questionnaire.

ANOVA? No categorical factors.

T-test? No group(s) to compare.

Correlation? Yes, because we are looking for a linear relationship between two continuous variables.

Examples of when ANOVAs are inappropriate

Research goal: Test whether RSI (repetitive stress injury) is decreased for a group of meditators vs. non-meditators.

Variables: IV: group (meditate or not). DV: scores on an RSI index.

ANOVA? Only one factor with two levels.

T-test? Yes, because there are two groups to compare.

Correlation? No, because the IVs are discrete groups, not continuous variables.

The way of the ANOVA

<number>-way: The number of factors.

Examples:

One-way ANOVA: Determine the influence of day-of-week on iPhone purchases.

Two-way ANOVA: Determine the influences of day-of-week and gender (male, female) on iPhone purchases.

Repeated-measures ANOVA

rmANOVA: If at least one factor involves multiple measurements from the same individual.

Example:

Research question: Determine effects of snack type on mood.
Experiment: Volunteers eat chocolate for 2 days, potato chips for 2 days, and ice cream for 2 days (order randomized).

Balanced vs. unbalanced ANOVA

Balanced: The same number of data points in each cell.

Unbalanced: Different number of data points across cells.

This could happen because of data collection or cleaning.

Balanced ANOVA

Medication

A

B

placebo

Younger

20

20

20

Age group

Older

20

20

20

Balanced vs. unbalanced ANOVA

Balanced: The same number of data points in each cell.

Unbalanced: Different number of data points across cells.

This could happen because of data collection or cleaning.

Unbalanced ANOVA

Medication

A

B

placebo

Younger

20

23

21

Older

18

20

20

Age group

Dummy-coding variables

Dummy-coding: Converting categorical variables into numbers.
Best applied to two cases with numbers {0,1}.

Lingo: "Gender was entered into the ANOVA as a dummy-coded variable (male=0, female=1)."

Interpretation: The effect of the factor shows the change for the "1" variable compared to the "0" variable ("0" acts like a baseline).

Example: Main effect of gender on lipstick use indicates that women use lipstick more than men (cannot claim anything about men using lipstick independently of women).

ANOVA vs. MANOVA

ANOVA: Only one DV (as many IVs as appropriate)

MANOVA (multivariate ANOVA): Multiple DVs (as many IVs as appropriate)

Example: Effects of medication type and age on Covid-19 symptoms and total medical expenses.

Fixed vs. random effects ANOVAs

Fixed effects: The number of levels of a factor is fixed (e.g., home type: dorm room, apartment, house).

Random effects: The levels of a factor are random (continuous) in the population (e.g., age, salary, nurse).

Mixed effects: Some factors are fixed; others are random.

Fixed vs. random effects ANOVAs

Fixed effects: The number of levels of a factor is fixed.

Random effects: The factor is random in the population.

Mixed effects: Both fixed and random factors.

Example mixed-effects ANOVA: Determine whether subjective happiness is influenced by home type (fixed effects) and age (random factor).

Another example: Determine whether the experimenter (random factor) influences the effects of medication (fixed).

Assumptions of ANOVA

Independence: The data are sampled independently of each other in the population to which you want to generalize.

Normality: The “residuals” (unexplained variance after fitting the model) are normally (Gaussian) distributed.

Homogeneity of variance (a.k.a. heteroscedasticity!): Variance within each cell is (roughly) the same.

Nonparametric ANOVA alternatives

Kruskal-Wallis test (KW-ANOVA): Alternative to the one-way ANOVA on rank-transformed data.

Note: ANOVAs are generally robust to violations of the assumptions. Nonparametric ANOVAs are rarely used and may cause more of a headache than they are worth.

Sum of squares

In this video, you will learn

- The null hypotheses that ANOVAs test against.
- The math behind testing ANOVAs (part 1).
- The degrees of freedom for each variance component.

Competing hypothesis of ANOVA

The null hypothesis is that the means of all groups (all cells in the ANOVA table) are statistically indistinguishable.

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

The alternative hypothesis is that the mean of at least one group is different from the mean of at least one other group.

$$H_A : \mu_i \neq \mu_j$$

Computing and interpreting sum of squares

$$SS = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

ANOVA as a partition of sum of squares

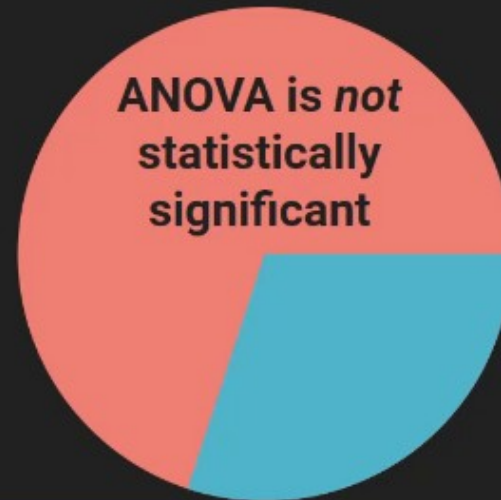
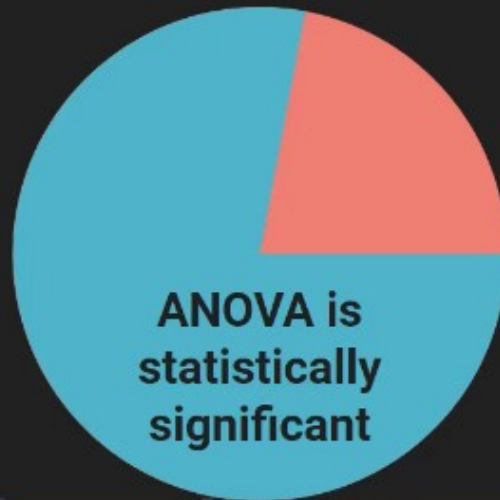
The total variation in the dataset is the sum of the variation across individuals within each group AND the variation across the different levels.

Total SS = Within-group SS + Between-group SS

Total SS = Error SS + Between-group SS

ANOVA as a partition of sum of squares

$$F = \frac{\text{"Explained" variance}}{\text{"Unexplained" variance}} = \frac{\text{Due to factors}}{\text{Natural variation}}$$



ANOVA as a partition of sum of squares

$$SS_{\text{Total}} = \sum_{j=1}^{\text{levels}} \sum_{i=1}^{\text{individuals}} (x_{ij} - \bar{x})^2 \quad df_{\text{Total}} = N - 1$$

$$SS_{\text{Between}} = \sum_{j=1}^{\text{levels}} (\bar{x}_j - \bar{x})^2 n_j \quad df_{\text{Between}} = k - 1$$

$$SS_{\text{Within}} = \sum_{j=1}^{\text{levels}} \sum_{i=1}^{\text{individuals}} (x_{ij} - \bar{x}_j)^2 \quad df_{\text{Within}} = N - k$$

Sum of squares and one-way ANOVA

Thus far, the explanations are for a one-way ANOVA.

This is for simplicity; adaptation to multi-way ANOVAs will be discussed in a few videos...

The F-test and the ANOVA table

In this video, you will learn

- How to average the sum of squares.
- How to compute F statistics and their p-values.
- How to create the ANOVA table.

Sum of squares to mean square

$$MS_{\text{Between}} = \frac{SS_{\text{Between}}}{df_{\text{Between}}} = \frac{\sum_{j=1}^{\text{levels}} (\bar{x}_j - \bar{x})^2 n_j}{k - 1}$$

$$MS_{\text{Within}} = \frac{SS_{\text{Within}}}{df_{\text{Within}}} = \frac{\sum_{j=1}^{\text{levels}} \sum_{i=1}^{\text{individuals}} (x_{ij} - \bar{x}_j)^2}{N - k}$$

The F-test

$$F_{k-1, N-k} = \frac{MS_{\text{Between}}}{MS_{\text{Within}}}$$

The ANOVA table

Source of variance	Sums of squares	Degrees of freedom	Mean square	F	P-value
Between groups	SS_B	$k-1$	MS_B	MS_B/MS_W	p
Within groups	SS_W	$N-k$	MS_W		
Total	SS_T	$N-1$			

Remember:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$
$$H_A : \mu_i \neq \mu_j$$

The ANOVA table

Source of variance	Sums of squares	Degrees of freedom	Mean square	F	P-value
Between groups	SS_B	$k-1$	MS_B	MS_B/MS_W	p
Within groups	SS_W	$N-k$	MS_W		
Total	SS_T	$N-1$			

Correct interpretation of $p < .05$: At least one level (group) is statistically significantly different from at least one other level. **Determining which groups differ requires data visualization follow-up t-tests.**

One-Way vs. Two-Way ANOVA

One-Way ANOVA

- Uses one independent variable or factor
- Assesses the impact of a single categorical variable on a continuous dependent variable, identifying significant differences among group means
- Does not account for interactions

Two-Way ANOVA

- Uses two independent variables or factors
- Used to not only understand the individual effects of two different factors, but also how the combination of these two factors influences the outcome
- Can test for interactions between factors

Cheat Sheet on Common Statistical Tests in Finance and Investing

Test	Purpose	When to Use	Applications in Finance/Investing
ANCOVA	Compares the arithmetical means of two or more groups while controlling for the effects of a continuous variable	• Normal distribution • Comparing multiple independent variables with a covariate	• Analyzing investment returns while controlling for market volatility • Evaluating the effectiveness of financial strategies while accounting for economic conditions
ANOVA	Compares the means of three or more groups	• Data is normally distributed	• Comparing financial performance across different sectors or investment strategies
Chi-Square Test	Tests for association between two categorical variables (can't be measured on a numerical scale)	• Data is categorical (e.g., investment choices, market segments)	• Analyzing customer demographics and portfolio allocations
Correlation	Measures the strength and direction of a linear relationship between two variables	• Data is continuous	• Assessing risk and return of assets, portfolio diversification

Cheat Sheet on Common Statistical Tests in Finance and Investing

Test	Purpose	When to Use	Applications in Finance/Investing
Durbin-Watson Test	Checks if errors in a prediction model are related over time	• Time series data	• Detecting serial correlation in stock prices, market trends
F-Test	Compares the variances of two or more groups	• Data is normally distributed	• Testing the equality of variances in stock returns and portfolio performance
Granger Causality Test	Tests for a causal relationship between two time series	• Time series data	• Determining if one economic indicator predicts another
Jarque-Bera Test	Tests for normality of data	• Continuous data	• Assessing if financial data follows a normal distribution

Cheat Sheet on Common Statistical Tests in Finance and Investing

Test	Purpose	When to Use	Applications in Finance/Investing
Mann-Whitney U Test	Compares medians of two independent samples	• Data is not normally distributed	• Comparing the financial performance of two groups with non-normal distributions
MANOVA	Compares means of two or more groups on multiple dependent variables simultaneously	• Data is normally distributed • Analyzing multiple related outcome variables	• Assessing the impact of different investment portfolios on multiple financial metrics • Evaluating the overall financial health of companies based on various performance indicators
One-Sample T-Test	Compares a sample mean to a known population mean	• Data is normally distributed, or the sample size is large	• Comparing actual versus expected returns
Paired T-Test	Compares means of two related samples (e.g., before and after measurements)	• Data is normally distributed, or the sample size is large	• Evaluating if a financial change has been effective

Cheat Sheet on Common Statistical Tests in Finance and Investing

Test	Purpose	When to Use	Applications in Finance/Investing
Regression	Predicts the value of one variable based on the value of another variable	• Data is continuous	• Modeling stock prices • Predicting future returns
Sign Test	Tests for differences in medians between two related samples	• Data is not normally distributed	• Non-parametric alternative to the paired t-test in financial studies
T-Test	Compares the means of two groups	• Data is normally distributed, or the sample size is large	• Comparing the performance of two investment strategies
Wilcoxon Rank-Sum Test	Compares the medians of two independent samples	• Data is not normally distributed	• Non-parametric alternative to the independent t-test in finance
Z-Test	Compares a sample mean to a known population mean	• Data is normally distributed, and the population standard deviation is known	• Testing hypotheses about market averages

The omnibus test and post-hoc comparison

In this video, you will learn

- What the “omnibus F-test” means.
- How to test pairs of levels in an ANOVA.
- The condition for when post-hoc comparisons are permitted.

The “problem” with the ANOVA F-test

	Sums of squares	Degrees of freedom	Mean square	F	P-value
Between groups	SS_B	$k-1$	MS_B	MS_B/MS_W	p
Within groups	SS_W	$N-k$	MS_W		
Total	SS_T	$N-1$			

Correct interpretation of $p < .05$: At least one level (group) is statistically significantly different from at least one other level. **Determining which groups differ requires data visualization follow-up t-tests.**

Example ANOVA table

	Sums of squares	Degrees of freedom	Mean square	F	P-value
Between groups	1850.47	3 (k-1)	616.82	13.25	.0002
Within groups	697.95	15 (N-k)	46.53		
Total	2548.42	18 (N-1)			

One-way ANOVA with four levels and 19 data points.

Example ANOVA table

	Sums of squares	Degrees of freedom	Mean square	F	P-value
Between groups	1850.47	3 (k-1)	616.82	13.25	.0002
Within groups	697.95	15 (N-k)			
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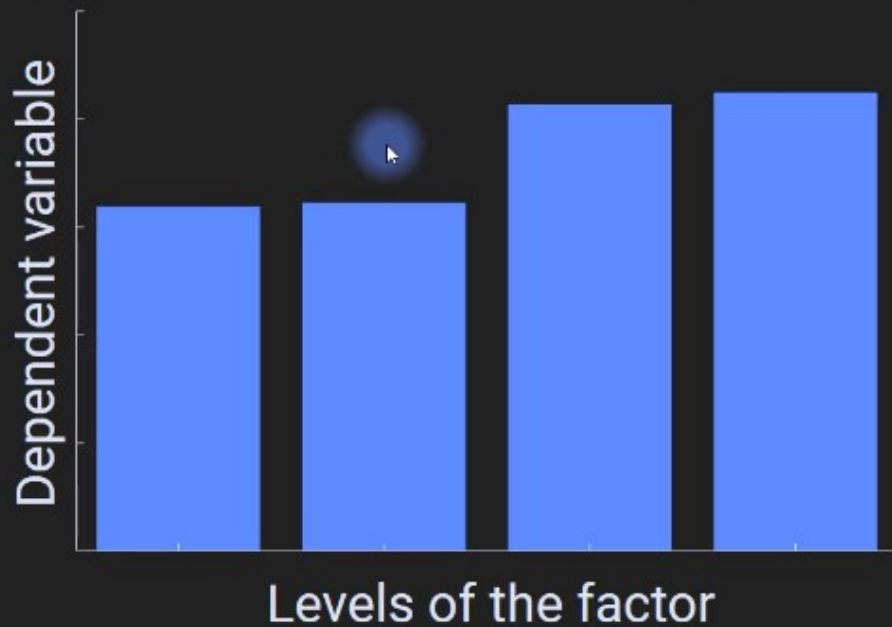
"Omnibus" F-test

One-way ANOVA with four levels and 19 data points.

Conclusion: The mean of at least one level is statistically significantly different from the mean of at least one other level.

Mo' comparisons, mo' problems

Problem: All possible comparisons (each at $p < .05$) leads to $6 * .05 = .3$ Type I error rate.



Thanks, Tukey, for the test

Solution: The Tukey test allows for post-hoc comparisons while controlling the familywise error rate.

Thanks, Tukey, for the test

Solution: The Tukey test allows for post-hoc comparisons while controlling the familywise error rate.

$$q = \frac{\bar{x}_b - \bar{x}_s}{\sqrt{MS_{\text{Within}}} \sqrt{2/n}}$$

q is evaluated with $(j, n-j)$ degrees of freedom.

j is the number of comparisons.
 n is the total number of data values.

Important note about post-hoc testing

Post-hoc comparisons within an ANOVA are allowed only when the omnibus F-test is significant.

No significant F-test -> no post-hoc comparisons.

Example ANOVA table

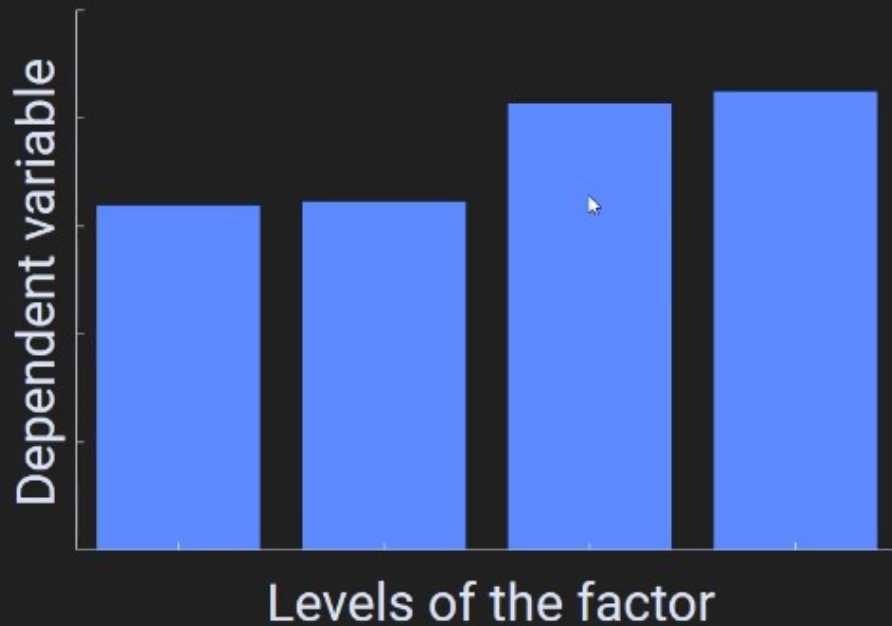
	Sums of squares	Degrees of freedom	Mean square	F	P-value
Between groups	1850.47	3 (k-1)	616.82	13.25	.0002
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One-way ANOVA with four levels and 19 data points.

Conclusion: The mean of at least one level is statistically significantly different from the mean of at least one other level.

Mo' comparisons, mo' problems

Problem: All possible comparisons (each at $p < .05$) leads to $6 * .05 = .3$ Type I error rate.



The two-way ANOVA

In this video, you will learn

- How to re-partition the total variance with more than one factor.
- The modifications to the SS formulas.
- How to create the 2-way ANOVA table.
- How to interpret main effects and interactions.

Extension to two-way ANOVA

Factors: Medication, age.

Levels: A, B, placebo (medication); younger, older (age)

		Medication		
		A	B	placebo
Age group	Younger			
	Older			

Partition of sum of squares in 2-way ANOVA

The total variation in the dataset is the sum of the variation across individuals within each group AND the variation across the different levels within each factor AND the variation at the interaction between the factors.

Extension to two-way ANOVA

$$SS_{\text{Total}} = \sum_{k=1}^{\text{levels B}} \sum_{j=1}^{\text{levels A}} \sum_{i=1}^{\text{individuals}} (x_{ijk} - \bar{x})^2$$

$$df_{\text{Total}} = N - 1$$

$$SS_{\text{BtwnA}} = bn \sum_{j=1}^{\text{levels A}} (\bar{x}_j - \bar{x})^2$$

$$df_{\text{BtwnA}} = a - 1$$

$$SS_{\text{BtwnB}} = an \sum_{k=1}^{\text{levels B}} (\bar{x}_k - \bar{x})^2$$

$$df_{\text{BtwnB}} = b - 1$$

$$SS_{\text{A} \times \text{B}} = \sum_{k=1}^{\text{levels B}} \sum_{j=1}^{\text{levels A}} (\bar{x}_{kj} - \bar{x}_k - \bar{x}_j + \bar{x})^2$$

$$df_{\text{A} \times \text{B}} = (a - 1)(b - 1)$$

$$SS_{\text{Within}} = \sum_{k=1}^{\text{levels B}} \sum_{j=1}^{\text{levels A}} \sum_{i=1}^{\text{individuals}} (x_{ijk} - \bar{x}_{jk})^2$$

$$df_{\text{Within}} = N - ab$$

Extension to two-way ANOVA

$$SS_{\text{BtwnA}} = bn \sum_{j=1}^{\text{levels A}} (\bar{x}_j - \bar{x})^2$$

$$SS_{\text{BtwnB}} = an \sum_{k=1}^{\text{levels B}} (\bar{x}_k - \bar{x})^2$$

$$SS_{\text{A} \times \text{B}} = n \sum_{k=1}^{\text{levels B}} \sum_{j=1}^{\text{levels A}} (\bar{x}_{kj} - \bar{x}_k - \bar{x}_j + \bar{x})^2$$

$$SS_{\text{A} \times \text{B}} = SS_{\text{Cell}} - SS_{\text{BtwnA}} - SS_{\text{BtwnB}}$$

The two-way ANOVA table

	Sums of squares	Degrees of freedom	Mean square	F	P-value
Factor A	SS_A	$a-1$	MS_A	MS_A/MS_W	p
Factor B	SS_B	$b-1$	MS_B	MS_B/MS_W	p
A×B interact.	$SS_{A \times B}$	$(a-1)(b-1)$	$MS_{A \times B}$	$MS_{A \times B}/MS_W$	p
Within (error)	SS_W	$N-ab$	MS_W		
Total	SS_T	$N-1$			

Correct interpretation of $p < .05$: At least one level (group) is statistically significantly different from at least one other level. **Determining which groups differ requires data visualization follow-up t-tests.**

Important take-home message

NEVER compute an ANOVA by hand!

How to interpret the main effects and interactions

A significant main effect means that the factor influences the data independently of the other factor(s).

A significant interaction means that the effect of one factor depends on another factor.

How to interpret the main effects and interactions

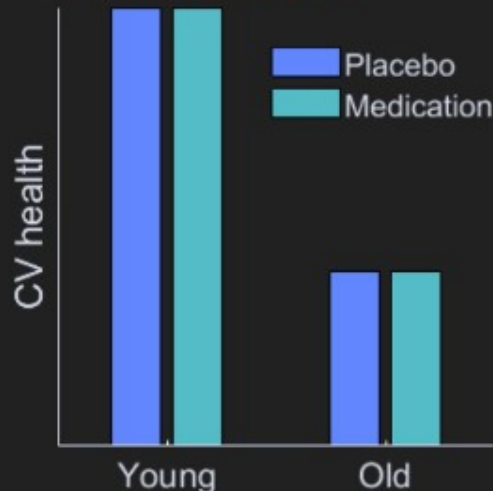
Example: The effects of vitamin supplement (vs. placebo) and age (30-40 vs. 50-60) on cardiovascular health.

Experiment design table:

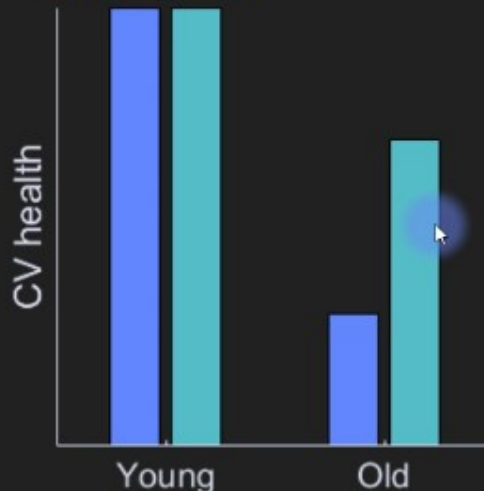
		Supplement	
		real	placebo
Age group	30-40	x	x
	50-60	x	x

How to interpret the main effects and interactions

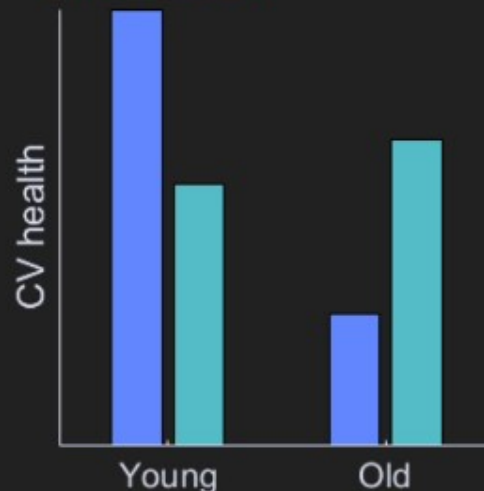
Main effect of age
No effect of medication
No interaction



Main effect of age
Effect of medication
Interaction



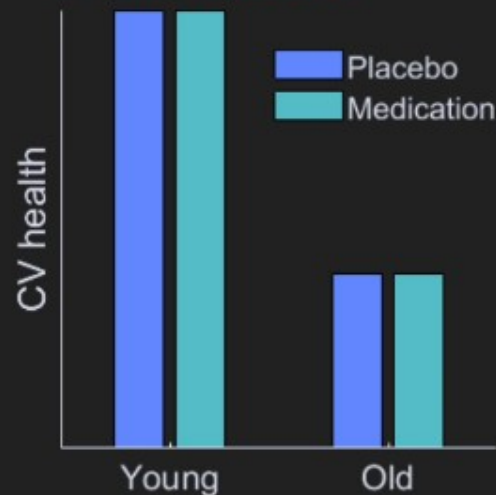
Main effect of age
No effect of medication
Interaction



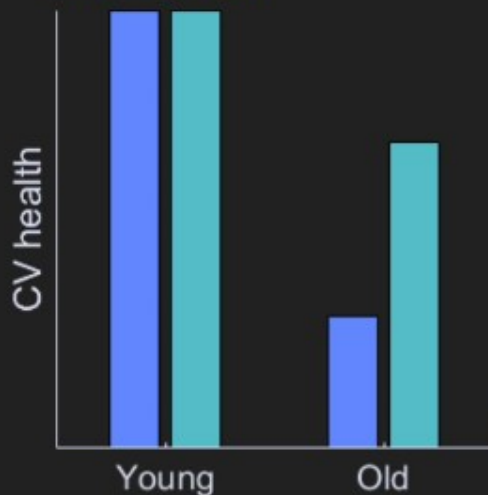
Important take-home message 1: Main effects must be cautiously interpreted with a significant interaction.

How to interpret the main effects and interactions

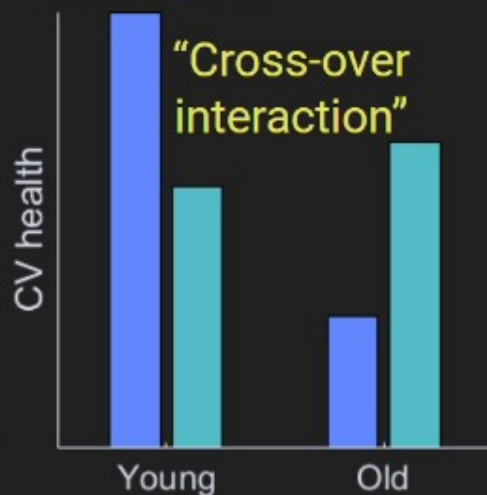
Main effect of age
No effect of medication
No interaction



Main effect of age
Effect of medication
Interaction



Main effect of age
No effect of medication
Interaction



Important take-home message 2: Data must always be visualized for proper interpretation!

One-way ANOVA: example

In this video, you will

- See an example of a one-way ANOVA.
- Gain more experience interpreting ANOVA results.

The experiment

Research goal: Test self-reported happiness after watching different genres of movies.

Variables: IV: movie genre (horror, romcom, documentary, sci-fi).
DV: happiness rating (1-100).

	Movie genre			
	Horror	Romcom	Docu	Sci-fi
N per group	5	5	4	5

The data

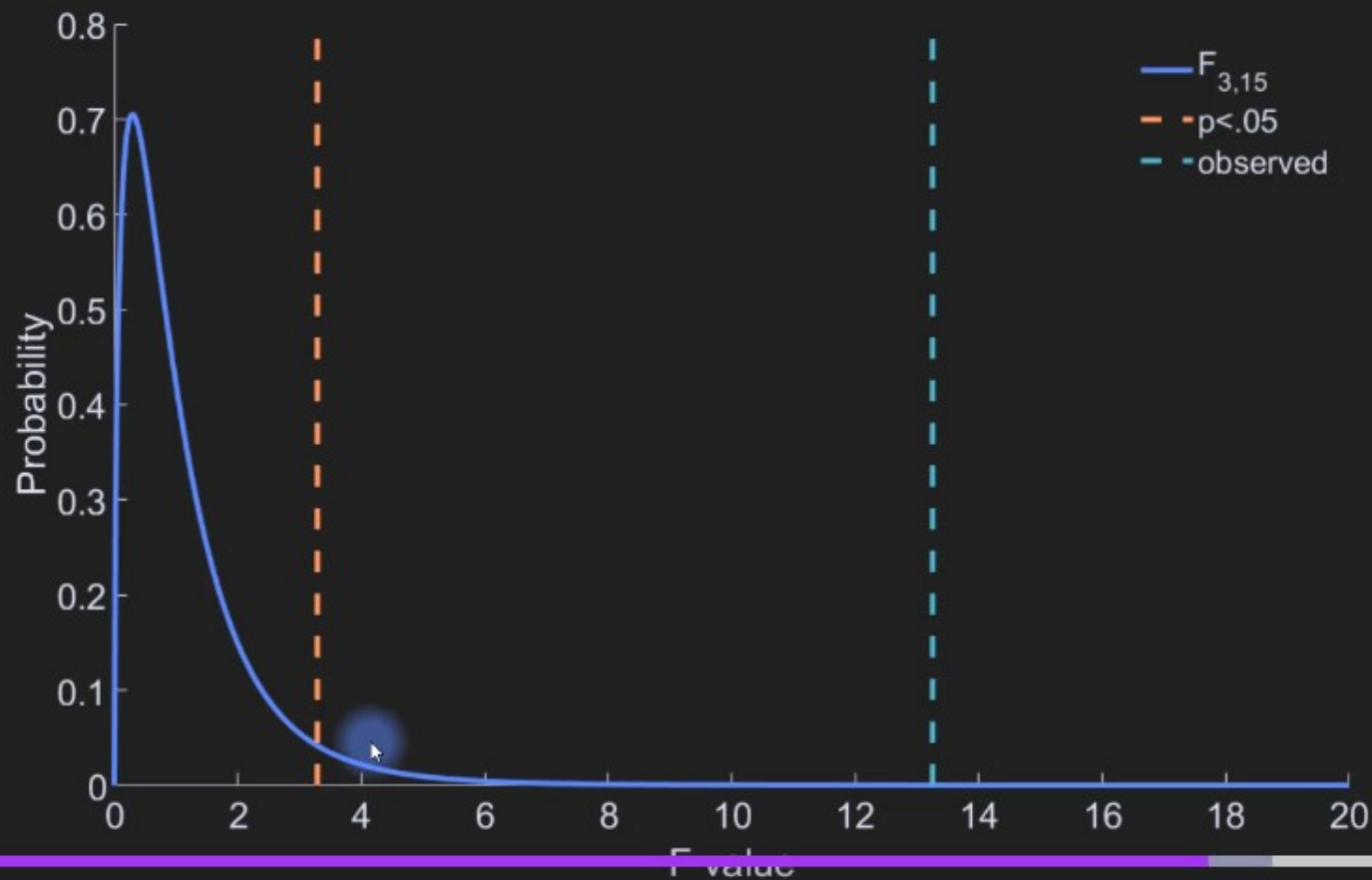
Happiness rating	Movie genre			
	Horror	Romcom	Docum.	Sci-fi
	61	68	82	90
	57	78	84	87
	70	65	90	75
	65	57	75	88
	67	55		85

* Note: Made-up data!

The ANOVA table

	Sums of squares	Degrees of freedom	Mean square	F	P-value
Between groups	1850.47	3 (k-1)	616.82	13.25	.0002
Within groups	697.95	15 (N-k)	46.53		
Total	2548.42	18 (N-1)			

The H_0 and observed F values



The ANOVA table

	Sums of squares	Degrees of freedom	Mean square	F	P-value
Between groups	1850.47	3 (k-1)	616.82	13.25	.0002
Within groups	697.95	15 (N-k)	46.53		
Total	2548.42	18 (N-1)			

SLAYT : 44

Conclusion: The mean of at least one level is statistically significantly different from the mean of at least one other level.

People were happier after at least one genre compared to at least one other genre.

The graphed results



Problem: Which levels (genres) are statistically significantly different from which others?

Solution: Test all pairs using Tukey post-hoc test.

Thus, the calculation is different. In the TukeyHSD, the q-statistics is based on the mean difference divide by the pool standard error of the sum of the means

$$q_s = \frac{Y_A - Y_B}{SE},$$

where Y_A is the larger of the two means being compared, Y_B is the smaller of the two means being compared, and SE is the standard error of the sum of the means **from all groups**.

If you perform the t-test, you use the pool standard deviation **only between 2 groups**.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{2}{n}}}$$

```

import numpy as np
import matplotlib.pyplot as plt
import pingouin as pg
import pandas as pd

## data parameters
# group means
mean1 = 4
mean2 = 3.8
mean3 = 7

# samples per group
N1 = 30
N2 = 35
N3 = 29

# standard deviation (assume common across groups)
stdev = 2

## now to simulate the data
data1 = mean1 + np.random.randn(N1)*stdev
data2 = mean2 + np.random.randn(N2)*stdev
data3 = mean3 + np.random.randn(N3)*stdev

datacolumn = np.hstack((data1,data2,data3))

# group labels
groups = ['1']*N1 + ['2']*N2 + ['3']*N3

# convert to a pandas dataframe
df = pd.DataFrame({'TheData':datacolumn,'Group':groups})

anova=pg.anova(data=df,dv='TheData',between='Group')
print("ANOVA:")
print(anova)
print("\n\n\n")
tukey=pg.pairwise_tukey(data=df,dv='TheData',between='Group')
print("TUKEY:")
print(tukey)

df.boxplot('TheData',by='Group');

plt.show()
~

```

```

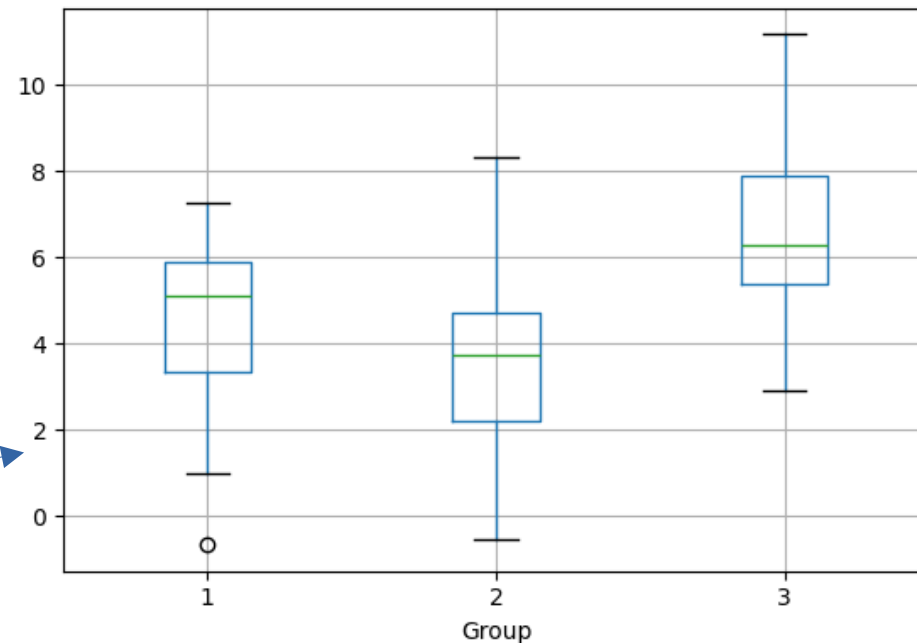
from pandascore import *

ANOVA:
  Source  ddof1  ddof2      F      p_unc      np2
0  Group      2     91  17.98176  2.624164e-07  0.283259

TUKEY:
   A  B  mean_A  mean_B  diff      se      T      p_tukey  hedges
0  1  2  4.548879  3.821960  0.726919  0.486499  1.494183  2.984762e-01  0.362618
1  1  3  4.548879  6.693423 -2.144544  0.509198 -4.211612  1.747269e-04 -1.112258
2  2  3  3.821960  6.693423 -2.871463  0.490995 -5.848251  2.300717e-07 -1.434726

```

Boxplot grouped by Group
TheData



Hipotez Testlerinde Pingouin Kullanımı: Veri Analizi ve Python
Emrecañ Ulu – Medium. (PDF olarak github'a konulmuştur.)

```

import numpy as np
import matplotlib.pyplot as plt
import pingouin as pg
import pandas as pd

## data parameters
# group means
mean1 = 4
mean2 = 3.8
mean3 = 7

# samples (same across group)
N = 30

# standard deviation (assume common across groups)
stdev = 2

## now to simulate the data
data1 = mean1 + np.random.randn(N)*stdev
data2 = mean2 + np.random.randn(N)*stdev
data3 = mean3 + np.random.randn(N)*stdev

datamat = np.vstack((data1,data2,data3)).T

# convert to a pandas dataframe
df = pd.DataFrame(data=datamat,columns=['d1','d2','d3'])
df

rm_anova_1=pg.rm_anova(data=df,detailed=True)
print(f"rm_anova_1 : \n {rm_anova_1}\n\n")
df.boxplot();
plt.show()

## example from SPSS website
# https://www.spss-tutorials.com/repeated-measures-anova/

data = [
    [8, 7, 6, 7],
    [5, 8, 5, 6],
    [6, 5, 3, 4],
    [6, 6, 7, 3],
    [8, 10, 8, 6],
    [6, 5, 6, 3],
    [6, 5, 2, 3],
    [9, 9, 9, 6],
    [5, 4, 3, 7],
    [7, 6, 6, 5]]

df = pd.DataFrame(data=data,columns=['1','2','3','4'])

rm_anova_2=pg.rm_anova(data=df,detailed=True)
print(f"rm_anova_2 : \n {rm_anova_2} \n\n")

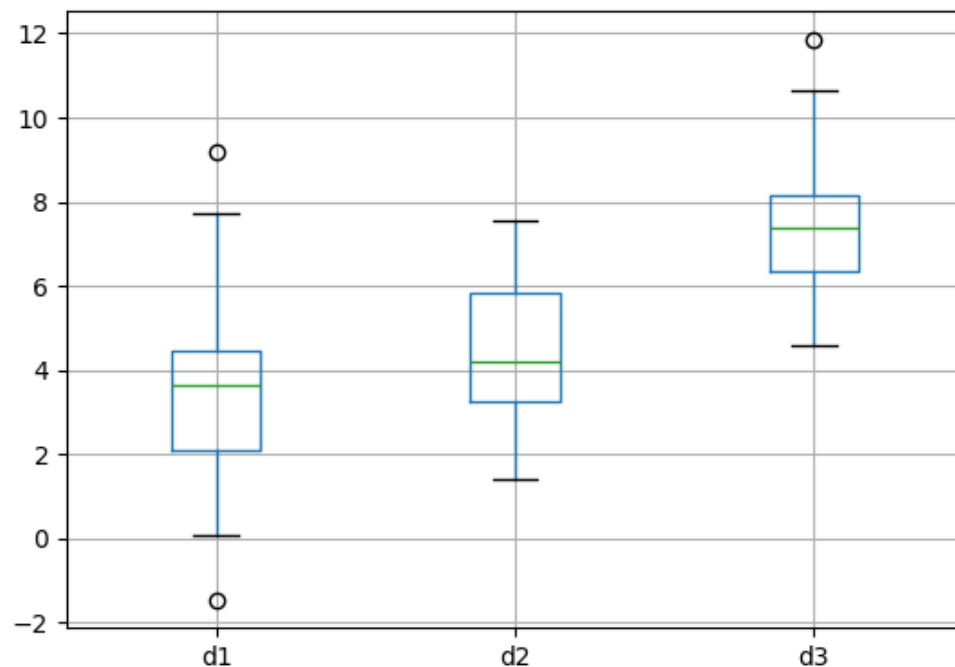
```

```

from pandas.core import (
rm_anova_1 :
  Source      SS    DF      MS      F      p_unc      ng2      eps
0  Within  230.518046    2  115.259023  30.310857  9.742047e-10  0.423886  0.910172
1   Error  220.548808   58    3.802566      NaN      NaN      NaN      NaN

rm_anova_2 :
  Source      SS    DF      MS      F      p_unc      ng2      eps
0  Within   18.2     3  6.066667  3.615894  0.025787  0.130372  0.697356
1   Error   45.3    27  1.677778      NaN      NaN      NaN      NaN

```



Two-way ANOVA: example

In this video, you will

- See an example of a two-way ANOVA.
- Gain more experience interpreting ANOVA results.
- Understand the importance of proper visualization for interpretation.

Experiment: Are girls bad at math?

Research goal: Test whether girls differ from boys in STEM and non-STEM subjects.

Variables: IV: class gender (boys, girls, mixed-boys, mixed-girls), subject (math, history).
DV: exam scores.

Experiment design: Three classes of 20 students each study the same material from the same teacher. An exam is given after two months.

**note: made-up numbers but conclusions are based on actual research.*

The ANOVA table

Source of variance	Sums of squares	Degrees of freedom	Mean square	F	P-value
Factor group					
Factor subj.					
G×S interact.					
Within (error)					
Total					

The ANOVA table

Source of variance	Sums of squares	Degrees of freedom	Mean square	F	P-value
Factor group		$a-1$			
Factor subj.		$b-1$			
G×S interact.		$(a-1)(b-1)$			
Within (error)		$N-ab$			
Total		$N-1$			

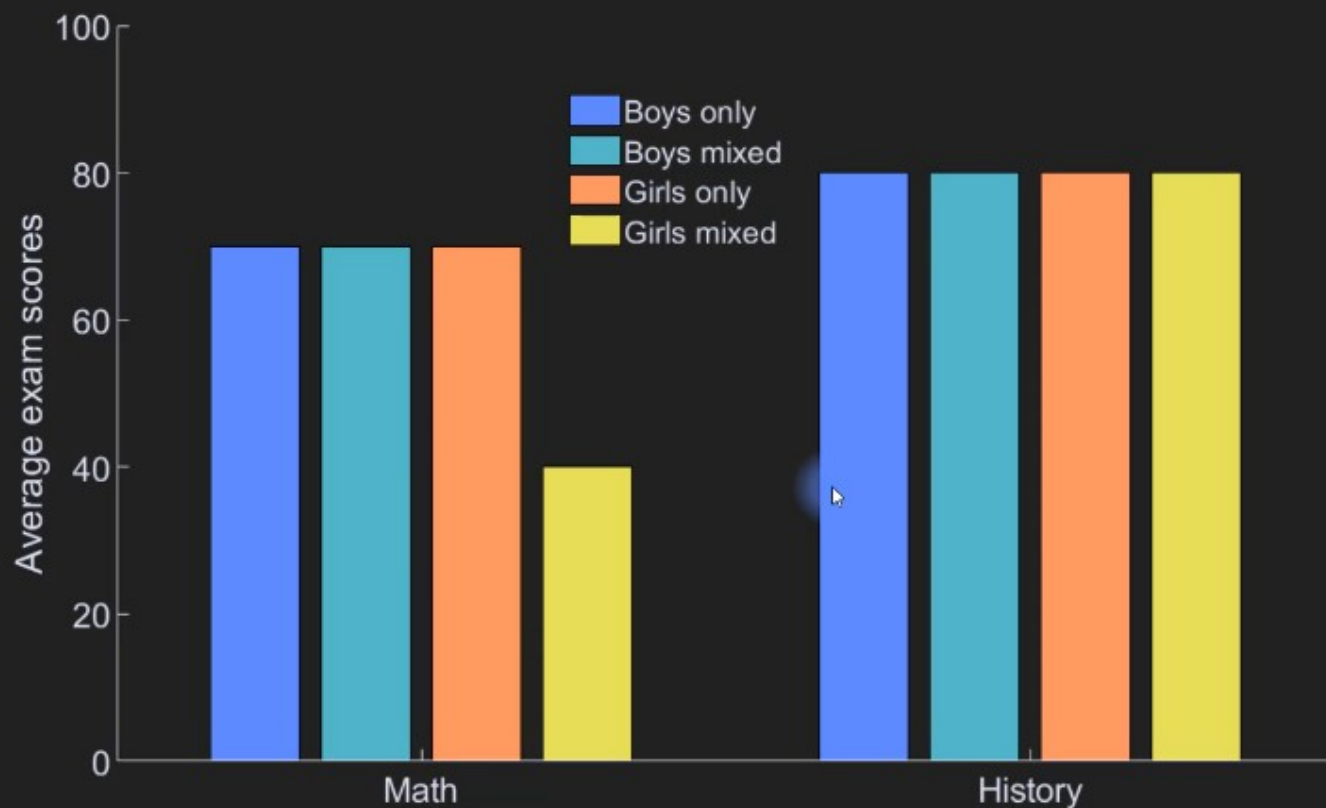
For a mixed-effects ANOVA (e.g., next video), the Total df is $(N-b)(a-1)$ where a is the within-subjects factor.

The ANOVA table

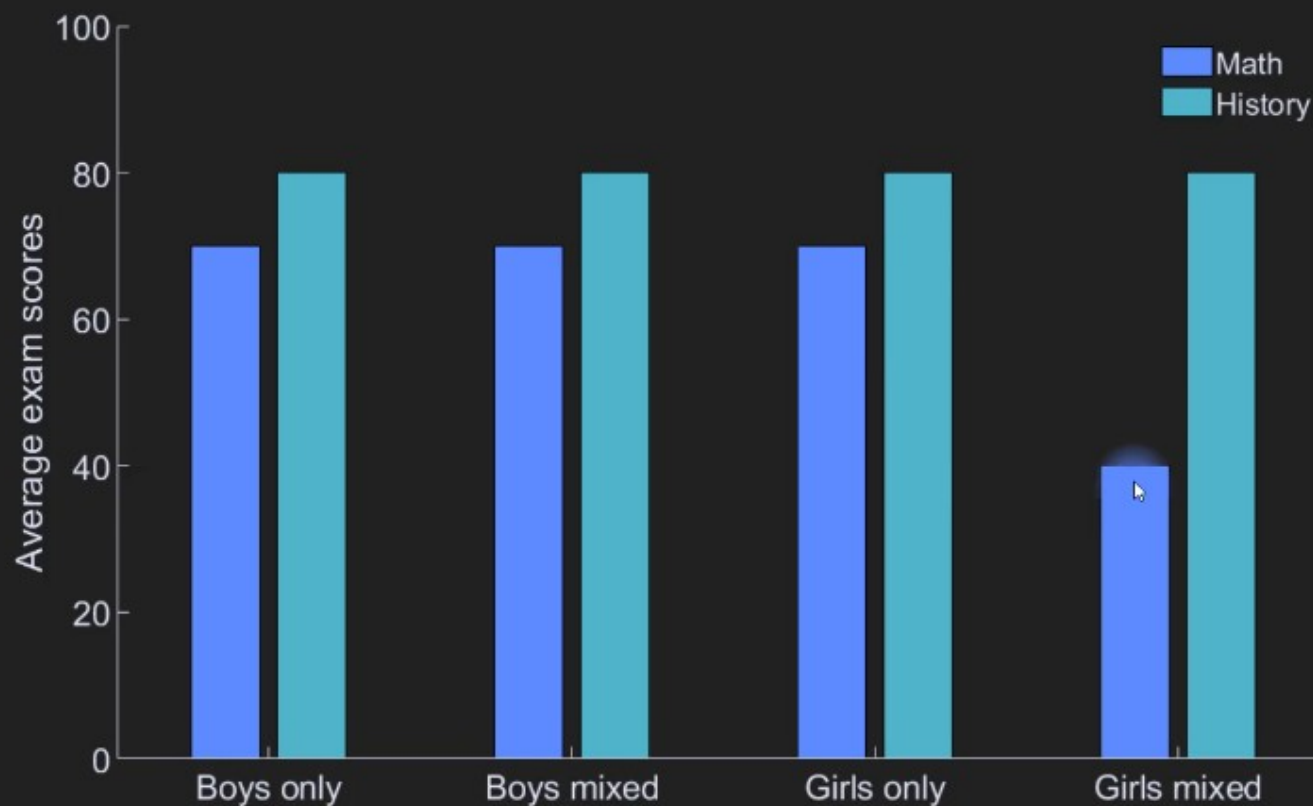
Source of variance	Sums of squares	Degrees of freedom	Mean square	F	P-value
Factor group		3 (4-1)			.456
Factor subj.		1 (2-1)			.028
G×S interact.		3 (4-1)(2-1)			.018
Within (error)		52 (60-4*2)			
Total		59 (60-1)			

Conclusion: There is a statistically significant interaction between Group and *subject*. The interaction makes the main effect of **subject** difficult to interpret without subsequent visualization and possible post-hoc tests.

Results, graphed



Results, graphed in a different way



```

import numpy as np
import matplotlib.pyplot as plt
import pingouin as pg
import pandas as pd
import seaborn as sns

# In[ ]:
## the data and group labels
data = np.loadtxt(open("TwoWayMixedANOVA_data.csv"),delimiter=",")

timepoint = ['1']*45 + ['2']*45 + ['3']*45
groups      = ['1']*15 + ['2']*15 + ['3']*15
s = []
for i in range(45):
    s += [str(i)]

# # convert to pandas
df = pd.DataFrame(data=np.matrix.flatten(data,'F'),columns=['TheData'])
df['Group'] = np.tile(groups,3)
df['TimePoint'] = timepoint
df['Subject'] = np.tile(s,3)

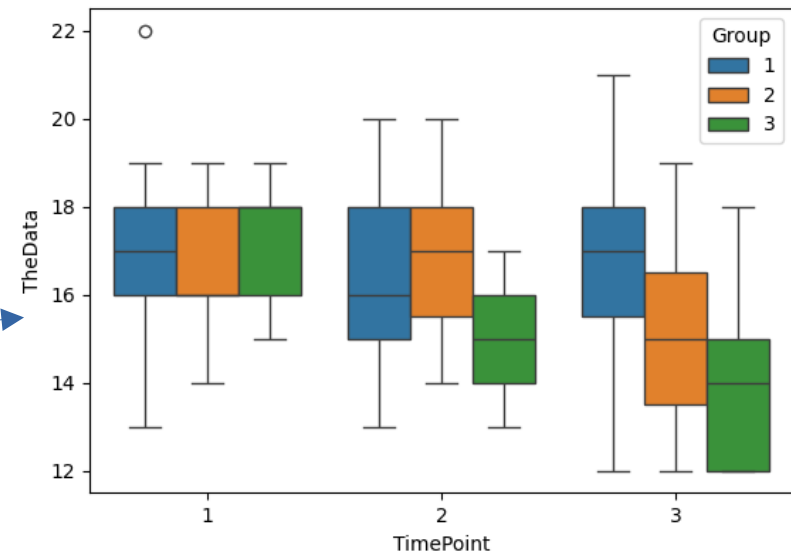
pd.set_option("display.max_rows", None, "display.max_columns", None)
df

# In[ ]:
pg.mixed_anova(data=df,dv='TheData',between='Group',within='TimePoint',sub

# In[ ]:
sns.boxplot(data=df,hue="Group",y="TheData",x='TimePoint')

plt.show()

```



```

13,14,12
15,13,15
13,14,15
16,15,13
17,16,16
18,18,18
16,15,17
16,16,18
18,18,17
17,17,18
17,18,16
19,16,19
18,18,21
19,19,17
22,20,19
14,14,12
16,18,13
15,14,13
16,15,14
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16,17,16
17,18,17
17,18,15
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18,18,16
18,20,16
18,19,17
18,19,19
19,17,19
16,14,12
15,15,12
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16,16,14
18,15,12
18,16,15
19,16,12
18,13,14
18,17,16
18,16,14
18,16,15
19,15,15
18,17,18

```